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Project 1: Drug Effectiveness Analysis

Objective

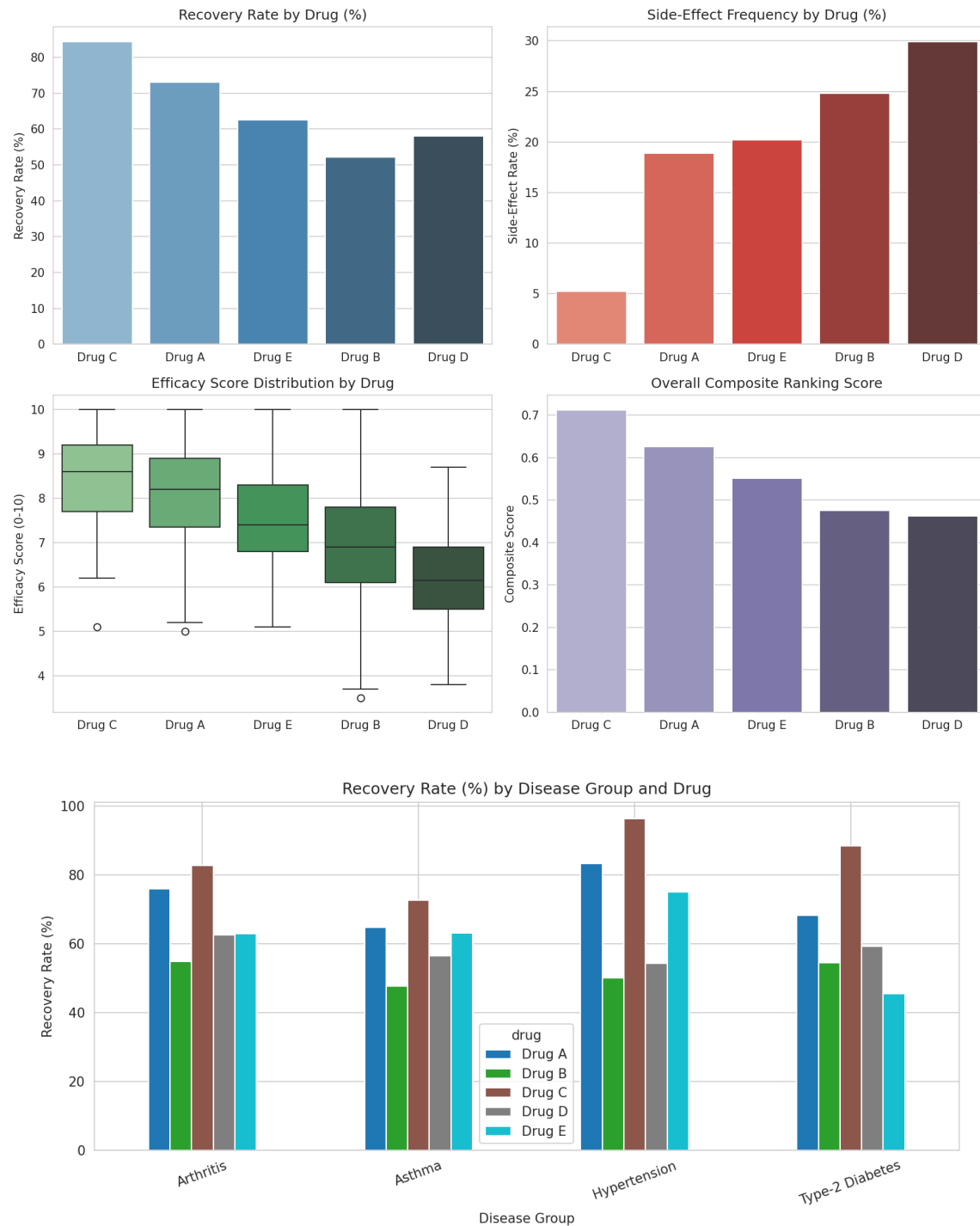
To evaluate and compare the performance of five drugs across 560 patient records, using recovery rate, side-effect frequency, and efficacy score to identify the most suitable drug candidate through a data-driven ranking system.

Approach

- Simulated a realistic patient dataset (560 records) with intentional data-quality issues (missing efficacy scores, duplicate entries, inconsistent gender labels).
- Cleaned the data using Pandas: removed duplicates, standardized categorical text, and imputed missing efficacy scores using drug-wise medians.
- Used groupby aggregation to compute recovery rate, side-effect rate, and average efficacy score for each drug.
- Built a composite ranking score:  $(\text{Recovery Rate} \times 0.45) + (\text{Efficacy}/10 \times 0.4) - (\text{Side-Effect Rate} \times 0.15)$ , balancing effectiveness against safety.
- Visualized results using Matplotlib/Seaborn: bar charts, box plots, and a disease-group breakdown.

Key Results

| Drug   | Patients | Recovery Rate | Side-Effect Rate | Avg Efficacy | Rank |
|--------|----------|---------------|------------------|--------------|------|
| Drug C | 115      | 84.3%         | 5.2%             | 8.50 / 10    | 1    |
| Drug A | 111      | 73.0%         | 10.8%            | 8.09 / 10    | 2    |
| Drug E | 104      | 62.5%         | 15.4%            | 7.42 / 10    | 3    |
| Drug B | 113      | 52.2%         | 23.9%            | 6.86 / 10    | 4    |
| Drug D | 117      | 58.1%         | 29.9%            | 6.11 / 10    | 5    |



## Outcome & Recommendation

Drug C emerged as the top-performing candidate, combining the highest recovery rate (84.3%) with the lowest side-effect frequency (5.2%) and the highest average efficacy score (8.50/10). This data-driven ranking approach reduces manual evaluation effort and gives decision-makers a clear, defensible basis for drug selection.

Tools used: Python (Pandas, NumPy, Matplotlib, Seaborn).